QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

umhackathon@um.edu.my

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Email: umhackathon@um.edu.my

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Document Control

Field	Detail
System Under Test (SUT)	*UMHackathon 2026 - Team MergeConflictSurvivor*
Team Repo URL	*https://github.com/yusufriduan/commission_compass*
Project Board URL	*https://commissioncompass.atlassian.net/jira/software/projects/COM/list?atlOrigin=eyJpIjoiNmU2OTNkZmEwNDA5NDdjNGI1MDI4YTUzY2Q2ZGFjNzYiLCJwIjoiaiJ9*

Objective:

Commission Compass aims to achieve a simple and accessible method for users to seek financial advice specifically tailored for the freelance industry. The system should be designed to enable users to easily make competent decisions specific to their situation by offering advice on pricing strategies, business planning, and managing income.

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

*This section aligns the testing connected back to specific user requirements via **Requirement Traceability Matrix".** So, it ensures every test has been conducted to prevent bugs/failure in products for that specific requirements and unplanned feature creep.*

1.1 In-Scope Core Features

Front-end implementation:

- **User Input** : Users will input specific details about their freelance work including the scope of the work, length of the work, urgency of the work's completion and the hourly rate for the work.

Back-end Implementation:

- **Personalized Advice Generation**: The system processes the user input through a deliberate structured pipeline and generates personalised financial advice. Firstly, the user input will be taken apart and analysed to determine key points and context such as the specific industry, risk level, and pricing. These highlighted points will then be fed into the language model, (ILMU-GLM-5.1), producing a structured output that is thoroughly validated ensuring consistency in responses and quality.

- **Conversation Tracking**: Conversations between the user and the AI will have a session-based memory, allowing the AI to have a more accurate context of the user's situation.

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1.2 Out-of-Scope

- **Processing images:** The application does not allow image processing and is only limited to text.
- **Regular conversations:** The application is only suited for direct professional advice, and will not converse casually with the user.
- **Project that involves multiple people working on the same task:** The application is only suitable to calculate the complexity of a commission task for a singular person instead of evaluating in context of multiple people.

2. Risk Assessment & Mitigation Strategy

*[e.g. Quality Assurance risks are anticipatory. So, identify the technical risk associated with your application requirements, architecture. Now, the risk needs to be evaluated using the 5*5 Risk assessment Matrix. So, Risk Score = Likelihood * Severity.]*

Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
Accidental non-inclusion of key factors of the commission (type of commission, deadline etc)	*3*	*4*	*12 (High)*	*The AI will prompt the user to key in the key factors of the commission to help finalise the system.*	*Input a commission for video editing without the key factor. Then afterwards key in said key factors.*

Risk Assessment Scoring Criteria:

Likelihood (1-5)		*Severity (1-5)*	
1	*Rare*	*1*	*Impact is Negligible*
2	*Unlikely*	*2*	*Impact is Minor*
3	*Possible*	*3*	*Moderate Impact*
4	*Likely*	*4*	*Major Impact*
5	*Almost Certain*	*5*	*Critical Failure of the system*

Risk Score = Likelihood × Severity

Risk Level Reference:

Risk Score	***Risk Level***	***Recommended Action***

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1 – 5	*Low*	*Monitor only. Acceptable risks.*
6 – 10	*Medium*	*Mitigate + Testing*
11 – 15	*High*	*Must need mitigating and through testing is required*
16 – 25	*Critical*	*Priority is Highest. Need extensive level of testing.*

3. Test Environment & Execution Strategy

Explain the testing environment. That means where the testing of your application will take place, how you are going to handle test data and the rules/threshold that you are incorporating to govern your testing phases.

- **Unit Test**
  - *Scope: Evaluation of commission's complexity score, obtaining the market benchmark,*
  - *Execution: The tests are done while using Docker Desktop*
  - *Isolation: To simply focus on the system's processing of the input and proper return of the output in json file format.*
  - *Pass Conditions: Achieved happy cases, provide negative cases with the proper error message and considered edge cases*
- **Integration Test**
  - *Determining the best course action for the input condition alongside a pros and cons list and suggestions*
  - *Execution: The test is performed after proper completion of the Unit Test to ensure the AI is proper using the tools given*
  - *Workflow: To ensure the API calls and interactions with databases are working, real calls are used for testing*
  - *Pass Condition: When the MCP tools within the AI are successfully used to obtain the final decision and shows a reasonable final decision*
- **Test Environment (CI/CD Practice):**
  - *Local Testing: Manual testing on localhost*
  - *Docker Desktop: Hosting images and containers on Docker Desktop to properly process the backend of the AI*

- **Regression Testing & Pass/Fail Rules:**
  - *The testing tests all of the AI's features every time a branch merge occurs to ensure the AI works as intended*
  - *Pass/Fail Condition: Test case pass is only considered when actual outcome perfectly aligns with expected one. If the requirements set for the test case is not fulfilled fully , the test status is considered a failure.*

- **Test Data Strategy:**
  - *Manual: creating 3 different scenarios considering freelancers of different job scopes and commission complexity.*

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- **Passing Rate Threshold:**
    - *A minimum of 85% of all test cases executed must pass. To avoid cost from critical case failure, all critical cases executed must pass.*

4. CI/CD Release Thresholds & Automation Gates

This section is mainly for defining metrics & threshold for success criteria. So, if a certain threshold is not met, the pipeline blocks the forward transition.

4.1 Integration Thresholds (Merging to Main)

The primary needs of this threshold to be crucial is because the main branch must be a stable source of truth. So, it needs to be achieved through continuous integration checks.

Checks	***Requirements***	***Project***	***Pass/Failed***
Automatic Build	***Zero Build Error***	***0 Errors***	***Passed***
Unit Tests	***100% Passing Rate***	***100%***	***Passed***
Code Quality	***Zero Linting Error***	***100%***	***Passed***
Test Coverage	***Minimum 81.2%***	***84%***	***Passed***

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4.2 Deployment Thresholds (Pushing to Production)

Checks	***Requirements***	***Project***	***Pass/Failed***
Regression Test	***Minimum 90%***	***91%***	***Passed***
AI Output Pass Rate	***Minimum of 80% of documented prompt***	***90%***	***Passed***
Critical Bugs	***Zero P0/01 Bugs***	***0***	***Passed***
API Performance	***Response Time < 800ms***	***450ms***	***Passed***
Security	***API keys are not exposed***	***Clean***	***Passed***

5. Test Case Specifications (Drafts)

This section is for you to draft your test cases. The test cases MUST include at least:

- ***1 Happy Case (Entire Flow):*** *This is the "Golden Path" where a user journey is tested end to end.*
- ***1 Specific Case (Negative/Edge Test):*** *This case should portray a scenario which triggers error in the system and the system handles it with proper handling techniques.*
- **2 Non-Functional (NFR) Tests (Performance/Load)*****:*** *This test consists of non-feature-based tests but architectural tests.*

Test Case ID	**Test Type & Mapped Feature**	**Test Description**	**Test Steps**	**Expected Result**	**Actual Result**
TC-01	***Happy Case (Entire Flow):*** *Vlog editing*	*Determine is the system can determine that*	*1. Open Website* *2. Key in the input about "getting a commission to edit a 2 hour footage into a full-length vlog*	*The commission is fully analysed by AI and AI comes to a final decision alongside key reasons, a list of pros and cons for the decision and*	*Successfully, Commission is fully analysed by AI and AI comes to a final decision alongside key reasonings, a list of pros and cons*

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
			in 3 hours. Should I accept the commission or should I increase the price?" *3. Obtain a decision*	*suggestions on the decision.*	*for the decision and suggestions on the decision.* *Status: Passed*
TC-02	***Specific Case (Negative):*** *Invalid Card Details*	*Verify the AI does handle error quite well when it receives data*	*1. Open Website* *2. Key in the input "what do you think about whales?"*	*AI detects unrelated topics and prompts the user to key in information regarding commissions.*	*AI detects unrelated topics and prompts the user to key in information regarding commissions.* *Status: Failed*
TC-03	***NFR (Performance):*** *API Response Test*	*API of the AI remains quite responsive (< 800ms) when concurrent users are using.*	*1. Open our app and select the API endpoint.* *2. 50 users GET Requests to that endpoint.* *3. Record the average response time.*	*Average response time < 800ms in addition 0% error rate.*	*Average Response time is received as 700ms. 0% Error Rate.* *Status: Pass*

6. AI Output & Boundary Testing (Drafts)

This section is designed to validate that your GLM integration does produce acceptable output and handles any inputs that are abnormal gracefully.

6.1. **Prompt/Response Test Pairs**

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Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
AI-01	I got a commission to edit 2 hours footage into a full length-vlog in 3 hours. Should I accept this commission or should I increase the price?	Provide a meaningful decision alongside good reasonings	decision": "Increase the price", "keyReasoningContent": "Editing 2 hours of raw footage into a cohesive vlog within a 3-hour deadline is a severe rush job. Just reviewing the footage at 1.5x speed takes over 1.3 hours, leaving less than 1.7 hours for the actual editing, color correction, audio balancing, and export. A standard vlog edit of this size typically takes 6-10 hours. The 3-hour constraint is unrealistic for quality work without cutting major corners, making a rush premium mandatory.", "prosList": ["Quick turnaround satisfies an urgent client need", "Rush jobs justify premium rates (1.5x to 2x your standard rate)", "Can lead to repeat business if you deliver under pressure"], "consList": ["Extremely high risk of burnout and subpar quality due to unrealistic timeline", "Just watching the footage consumes nearly half your allotted time", "No buffer for revisions, rendering errors, or client feedback", "High stress and potential damage to portfolio if the final product suffers"], "quantifiableImpactContent": "At an average market rate of $60/hr for vlog editing, a standard 8-hour edit would cost $480. A 3-hour rush deadline requires a 1.5x-2x multiplier, valuing the project at	Pass

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			$720-$960. If you accept the original flat rate without a rush fee, you are effectively devaluing your work to roughly $20-$30/hr while taking on extreme pressure.", "suggestionsContent": "Quote the client a rush fee of 1.5x to 2x your standard rate. Alternatively, negotiate the deadline: ask for 6-8 hours to deliver quality work at your standard rate. If the 3-hour deadline is non-negotiable, clearly define a 'no revisions' clause in the contract due to the time constraint, and require upfront payment before starting."	
AI-02	I got a commission for RM 100 for a full-body drawing commission. Should I accept this commission or should I increase the price?	Provide a meaningful decision alongside good reasonings, ensuring the commissioner gets their worth	"decision": "Counteroffer / Go Higher", "keyReasoningContent": "RM100 (approximately $21 USD) is generally considered very low for a full-body drawing commission. Even accounting for local economic factors, a full-body piece requires significant time, effort, and skill. Accepting this rate risks severe underpayment and sets a low precedent for your freelance business, unless the artwork is just a basic sketch.", "prosList": ["Quick to secure the job if the client is on a strict budget", "Good practice or portfolio builder if you are an absolute beginner", "Low client expectations regarding turnaround or revisions at this price point"], "consList": ["Severe underpayment for the hours typically required for a	Pass

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			full-body drawing", "Sets a low baseline that makes it hard to charge higher rates later", "High risk of burnout from working for far below minimum wage", "Attracts clients who undervalue artistic labor"], "quantifiableImpactContent": "At RM100, if the full-body drawing takes you 5 hours, your effective hourly rate is RM20/hour (~$4.20 USD). The global average for freelance illustration is around $60 USD/hour. Even for local Malaysian freelance rates, a standard colored full-body commission typically ranges from RM150 to RM400+, depending on rendering and detail. By accepting RM100, you are potentially losing out on RM50 to RM300+ of fair market value per piece.", "suggestionsContent": "1. Define the scope: Ask the client if RM100 is for a sketch, lineart, or fully rendered color. If it is fully rendered, politely counteroffer with your standard rate (e.g., RM250+). 2. Break down your pricing: Explain to the client what goes into a full-body drawing to justify your higher rate. 3. Offer tiers: If they only have RM100, offer a half-body or a simple sketch instead of a full-body drawing. 4. Limit revisions: If you do accept the RM100 for any reason, strictly limit the number of free revisions to avoid scope creep."	
AI-03	Should I take a job cinematic shoot	Provide a good decision while	"decision": "Decline the shoot or negotiate an earlier wrap	

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	for 3 days and 2 nights at pahang, even though I have OOPDS term test on the next day after I finished the shoot?	giving realistic reasonings and pros and con for the decision	time on the final day", "keyReasoningContent": "Taking a 3-day, 2-night intensive shoot immediately before a term test creates an extreme risk of academic failure and physical exhaustion. Travel fatigue from Pahang combined with the high cognitive and physical demands of a cinematic shoot leaves virtually no time or mental bandwidth to study or perform well on your OOPDS term test. Academic penalties usually outweigh a single freelance gig's benefits.", "prosList": ["Potential for a good portfolio piece and client relationship", "Earning income from a multi-day gig", "Gaining on-location cinematic shooting experience in Pahang"], "consList": ["Severe lack of preparation time for the OOPDS term test", "High risk of physical and mental exhaustion leading to poor exam performance", "Travel delays from Pahang could cause you to miss the test entirely", "Burnout risk from immediately transitioning from a high-stress shoot to a high-stress academic evaluation"], "quantifiableImpactContent": "Financial gain from a 3-day shoot (approx. $1,440 based on $60/hr standard, though actual lump sum may vary) must be	

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			weighed against the cost of failing the OOPDS term test. Failing could mean retaking the module, costing hundreds or thousands in tuition, plus the delayed graduation opportunity cost. A complexity score of 8.5/10 indicates high stress and physical demand, which practically guarantees zero productive study time on the last day of the shoot or the travel day back.", "suggestionsContent": "1. Decline the job politely, citing an unavoidable academic conflict, to protect your grades. 2. If you absolutely want the gig, negotiate with the client to wrap the shoot by early afternoon on the final day so you can travel back and study that evening. 3. Ask your professor if the term test can be taken on an alternative date (though this is rarely accommodated without a medical emergency). 4. If you decline, refer a trusted fellow filmmaker to the client to maintain a good professional relationship." }	

6.2. Oversized/Larger Input Test

In this section, do define the maximum input size your system accepts. Do highlight here what happens when the limit is exceeded.

Fields	Details
Maximum Input Size	400 words

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Input used while testing	500 words
Expected Behavior	Truncate the words within the input
Actual Behavior	Output appears as normal
Status	Pass

6.3. Adversarial/Edge Prompt Test

Fields	Details
Input used while testing	Mention about commission while sidetracking to a story about a whale
Expected Behavior	An Exception Error occurs
Actual Behavior	Output appears as the expected behaviour intends
Status	Pass

In this section, test minimum of 1 with one prompt that is ambiguous, malformed - document how GLM does respond and how your system is going to handle it.

6.4. Hallucinating Handling

This section is defined for you to describe how your system does detect or mitigate hallucinating response.

Fields	Details
Hallucinating Output	Forms a decision, but the reasoning does not relate to the intended decision
Expected Behavior	HTTP Exception error 500 will occur
Actual Behavior	Output appears as the expected behaviour intends
Status	Pass

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Important Note:

Using Agile principle, these testing will take place iteratively over the period of development, not limited to at the end.

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